

1D CNN — Architecture Variant Comparison

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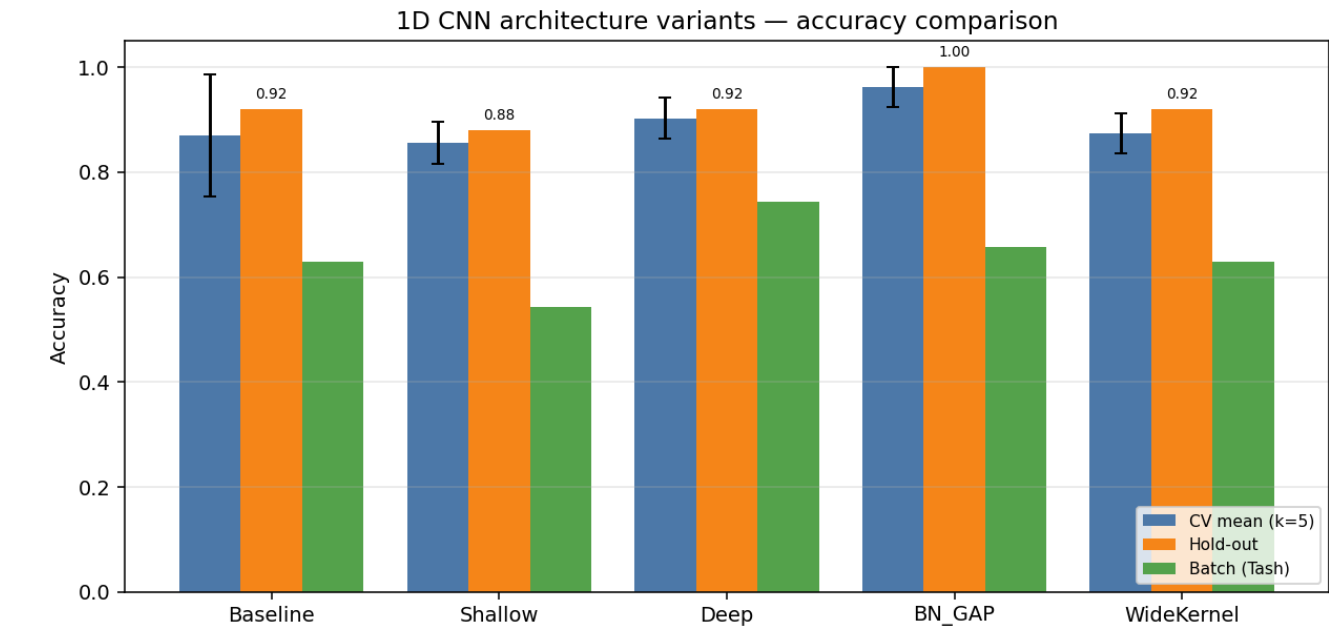
Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTestData/Alan/Dynamic
Training path 2	/home/jestin/ThesisRepo/ML/NewTestData/Anghad/Dynamic
Training path 3	/home/jestin/ThesisRepo/ML/NewTestData/Harry/Dynamic
Training path 4	/home/jestin/ThesisRepo/ML/NewTestData/Bella/Dynamic
Training path 5	/home/jestin/ThesisRepo/ML/NewTestData/Jestin/Dynamic
Training path 6	/home/jestin/ThesisRepo/ML/NewTestData/Joselyn/Dynamic
Training path 7	/home/jestin/ThesisRepo/ML/NewTestData/Marcus/Dynamic
Training path 8	/home/jestin/ThesisRepo/ML/NewTestData/Raquel/Dynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/Tash/Dynamic
Total trials	241 (train pool 216, hold-out 25)
Classes	7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	176 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 10.0 Hz, order 4
Normalisation	minmax
CV folds · shuffle	5 · True
Augmentation	on · copies=5 · amplitude_scale, baseline_offset
Epochs · batch size	15 · 16

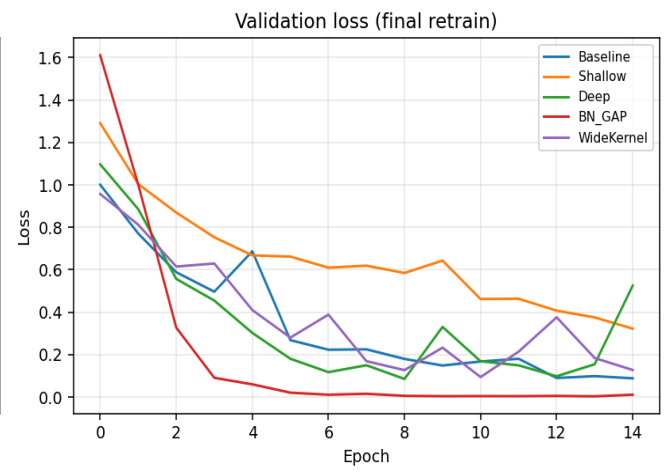
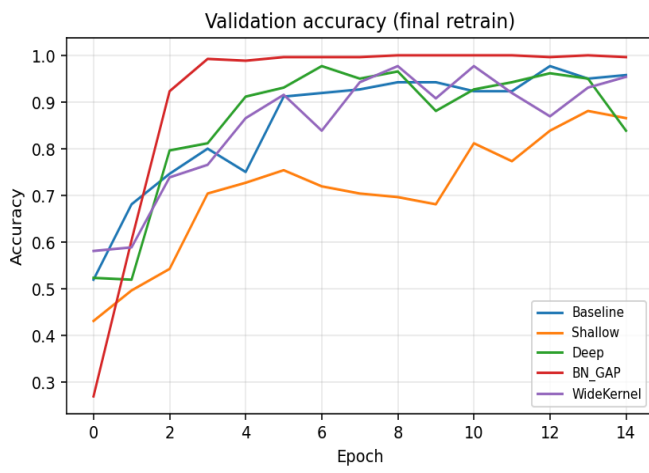
Variant comparison (summary)

Variant	Params	CV mean (k=5)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Baseline	113,255	0.8699	0.1165	0.9200	0.1880	0.6286	22/35
Shallow	72,487	0.8562	0.0406	0.8800	0.3575	0.5429	19/35
Deep	204,007	0.9029	0.0398	0.9200	0.2823	0.7429	26/35
BN_GAP	57,319	0.9628	0.0378	1.0000	0.0112	0.6571	23/35
WideKernel	131,687	0.8747	0.0383	0.9200	0.3190	0.6286	22/35

Best on hold-out: **BN_GAP** · Best on CV: **BN_GAP**



Training curves (final retrain on full training pool)



Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	113,255
CV mean ± std	0.8699 ± 0.1165
Hold-out test acc / loss	0.9200 · 0.1880
Batch-test acc	0.6286 (22/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1032	44	0.9773	0.9773	0.9773
2	1038	43	0.6512	0.9767	0.6512
3	1038	43	0.9535	0.9535	0.9535
4	1038	43	0.9070	0.9302	0.9070
5	1038	43	0.8605	0.8837	0.8605

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	4
Double_Nothing	1.000	0.333	0.500	3
Double_Pistol_Recoil	0.800	1.000	0.889	4
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	0.750	1.000	0.857	3
Double_cmere	1.000	1.000	1.000	4
Drum_Roll	1.000	1.000	1.000	3
macro avg	0.936	0.905	0.892	25
weighted avg	0.938	0.920	0.905	25

Batch test per class (Tash)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	0	5	0.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	2	5	0.400
Drum_Roll	0	5	0.000
Overall	22	35	0.629

Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

Trainable params	72,487
CV mean ± std	0.8562 ± 0.0406
Hold-out test acc / loss	0.8800 · 0.3575
Batch-test acc	0.5429 (19/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1032	44	0.9091	1.0000	0.9091
2	1038	43	0.8837	0.9535	0.8837
3	1038	43	0.7907	0.7907	0.7907
4	1038	43	0.8605	0.9302	0.8605
5	1038	43	0.8372	0.8605	0.8372

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	0.750	0.750	0.750	4
Double_Nothing	1.000	0.333	0.500	3
Double_Pistol_Recoil	0.800	1.000	0.889	4
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	0.750	1.000	0.857	3
Double_cmere	1.000	1.000	1.000	4
Drum_Roll	1.000	1.000	1.000	3
macro avg	0.900	0.869	0.857	25
weighted avg	0.898	0.880	0.865	25

Batch test per class (Tash)

Class	Correct	Total	Accuracy
Double_Lower	1	5	0.200
Double_Nothing	2	5	0.400
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	1	5	0.200
Drum_Roll	0	5	0.000
Overall	19	35	0.543

Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

Trainable params	204,007
CV mean ± std	0.9029 ± 0.0398
Hold-out test acc / loss	0.9200 · 0.2823
Batch-test acc	0.7429 (26/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1032	44	0.8864	0.9545	0.8864
2	1038	43	0.9767	1.0000	0.9767
3	1038	43	0.8837	0.9070	0.8837
4	1038	43	0.9070	0.9535	0.9070
5	1038	43	0.8605	0.8837	0.8605

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	0.800	1.000	0.889	4
Double_Nothing	0.667	0.667	0.667	3
Double_Pistol_Recoil	1.000	1.000	1.000	4
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	1.000	1.000	1.000	3
Double_cmere	1.000	1.000	1.000	4
Drum_Roll	1.000	0.667	0.800	3
macro avg	0.924	0.905	0.908	25
weighted avg	0.928	0.920	0.918	25

Batch test per class (Tash)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	1	5	0.200
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	0	5	0.000
Overall	26	35	0.743

Variant: BN_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

Trainable params	57,319
CV mean ± std	0.9628 ± 0.0378
Hold-out test acc / loss	1.0000 · 0.0112
Batch-test acc	0.6571 (23/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1032	44	1.0000	1.0000	1.0000
2	1038	43	0.9767	1.0000	0.9767
3	1038	43	1.0000	1.0000	1.0000
4	1038	43	0.9302	0.9535	0.9302
5	1038	43	0.9070	0.9070	0.9070

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	4
Double_Nothing	1.000	1.000	1.000	3
Double_Pistol_Recoil	1.000	1.000	1.000	4
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	1.000	1.000	1.000	3
Double_cmere	1.000	1.000	1.000	4
Drum_Roll	1.000	1.000	1.000	3
macro avg	1.000	1.000	1.000	25
weighted avg	1.000	1.000	1.000	25

Batch test per class (Tash)

Class	Correct	Total	Accuracy
Double_Lower	4	5	0.800
Double_Nothing	3	5	0.600
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	1	5	0.200
Drum_Roll	0	5	0.000
Overall	23	35	0.657

Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	131,687
CV mean ± std	0.8747 ± 0.0383
Hold-out test acc / loss	0.9200 · 0.3190
Batch-test acc	0.6286 (22/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1032	44	0.9318	0.9545	0.9318
2	1038	43	0.8605	0.9767	0.8605
3	1038	43	0.8837	0.9535	0.8837
4	1038	43	0.8837	0.8837	0.8837
5	1038	43	0.8140	0.8605	0.8140

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	0.800	1.000	0.889	4
Double_Nothing	1.000	0.333	0.500	3
Double_Pistol_Recoil	0.800	1.000	0.889	4
Double_Raise	1.000	1.000	1.000	4
Double_Wiggle	1.000	1.000	1.000	3
Double_cmere	1.000	1.000	1.000	4
Drum_Roll	1.000	1.000	1.000	3
macro avg	0.943	0.905	0.897	25
weighted avg	0.936	0.920	0.904	25

Batch test per class (Tash)

Class	Correct	Total	Accuracy
Double_Lower	4	5	0.800
Double_Nothing	0	5	0.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	3	5	0.600
Drum_Roll	0	5	0.000
Overall	22	35	0.629